

Task 1: Linux and Github

Linux Filesystem Hierarchy and Essential Commands:

Exploring different file system hierarchy in linux

/bin

```
ansh@diazolaptop:/bin$ cd /sbin
ansh@diazolaptop:/sbin$ ls
aa-load                  cups-genppdupdate      getty                   jfs_mkfs                mount.fuse               route                    update-mime
aa-remove-unknown        cupsreject              gnome-menus-blacklist kbdrate                 mount.fuse3              rtactt                  update-passwd
aa-status                dcb                     groupadd                key.dns_resolver        mount.lowntfs-3g         rtacct                  update-pciids
aa-teardown              debugfs                 groupdel                killall5                 mount.ntfs               rtcwake                  update-rc.d
accessdb                 debugreiserfs           groupmeas               kpartx                  mount.ntfs-3g            rtkitctl                update-secureboot-policy
addgnupghome             delgroup                groupmod                ldattach                 mount.smb3               rtmn                    update-shells
addgroup                 deluser                 grpck                   ldconfig                 nameif                    runlevel                 update-smart-drivedb
add-shell                depmod                  grproc                  ldmconfig                netplan                  runuser                  update-xmlcatalog
adduser                  devlink                 grpconv                 ldconf.real              NetworkManager           samba.kcc                upgrade-from-grub-legacy
agetty                   dhclient                grub-bios-setup         lightdm-session          newusers                  saned                    usb_modeswitch
alsa                     alsa-nb-test            dhcpcd                 logrotate                 nft                      select-default-ispell    usb_modeswitch_dispatcher
alsactl                  dkms                    grub-macless            logrotate                 nftcl                    select-default-wordlist  ushmrtd
alsa-info                dmeventd               grub-mkdevicemap        losetup                  nftsclo                   sensors-detect           useradd
apparmor_parser          dmidecode              grub-mkconfig           lpadmin                   ntfsclone                 service                  userdel
apparmor_status          dmraid                 grub-probe              lspci                     ntfslabel                 setcap                   usermod
appliednugdefaults       dmraid-activate        grub-reboot             lscap                     ntfsresize                setvesablink             uidd
apt                        dmsd                   grub-set-default        lshw                      ntfsundelete              setvtrgb                 validlocale
aptk                      dmsdts                 guest-account            lsmove                    on_ac_power              sfdisk                   vcsttime
arp                       dnsmaq                 halt                     lsmod                     openvpn                   sgdisk                   vdpd
arpd                      dosfsck                 hdparr                  lsppc                     ownershp                   shadowconfig             verifysetup
arptables                 dosfslabel             iconvconfig              lsppc                     pam-auth-update           shutdown                  vgcfgbackup
arptables-nft             dpkg-preconfigure      ifconfig                 lvs                       pam-extrousers_chkpw     slattach                 vgcfgrestore
arptables-nft-restore     dpkg-reconfigure       insmod                   lvchange                  pam-extrousers_update    slick-greeter            vgck
arptables-nft-save        dump2fs                 installkernel            lvconvert                 pam_getenv                smartctl                 vgconvert
arptables-restore         dump.exfat              install-smalcatalog      lvcreate                  pam_namespace_helper     start-stop-daemon        vgcreate
arptables-save            e2fsck                  integritysetup           lvdisplay                  pam_timestamp_check      sudo_logsrvd             vgdisk
aspell-autobuildhash      e2fsck                  invoke-rc.d              lvextend                   paperconfig               sudo_sendlog             vgexport
avahi-autoipd             e2image                 ip                       lvmconfig                 parted                    sulogin                  vgstend
avahi-daemon              e2label                 ip6tables                lvmformat                 pccardctl                 swaplabel                 vgmport
badblocks                 e2mmpstatus            ip6tables-apply          luksformat                 pccardctl                 swapoff                   vgmportclone
biosdecode                e2scrub                 ip6tables-legacy         lvchange                   pivot_root                swapon                    vmmerge
blkdeactivate             e2scrub_all             ip6tables-legacy-restore lvmpoll                    plipconfig                 switch_root               vgmnodes
blkdiscard                e2undo                  ip6tables-legacy-save    lvmsadc                    plocate-build             sync-avaiable            vgreduce
blkid                     e4crypt                 ip6tables-nft            lvmsar                     plymouth                   sysctl                    vgrename
blkzone                   e4defrag               ip6tables-nft-restore    lvreduce                    pm-hibernate              system-tools-backends     vgs
blockdev                  ebttables               ip6tables-nft-save       lvremove                    pm-powersave             tarcat                    vgs
bluetooth                 ebttables-nft           ip6tables-restore        lvrmoe                     pm-suspend                tc                         tcptraceroute             vgsplit
bpftool                   ebttables-nft-restore   ip6tables-restore-translate lvs                         pm-suspend-hybrid         tcpraceroute              vigr
bridge                     ebttables-nft-save      ip6tables-save            lvsan                      poweroff                   tcpraceroute.db          vipw
brltty                     ebttables-restore       ip6tables-translate      lvsan                      pppd                       tcl                         thermald                   visudo
brltty.setup              ebttables-save          ip6tables-translate       mdadm                      pppoe-discovery           tips                      traceoute                  vpdcode
capsh                      ebttables-translate     ippevepcl                mdmon                      pppstats                   tune2fs                  traceroute                  wipefs
casper-getty              exfat2img               ippeveprinter            mii-tool                  pptp                        tune2fs                   wpa_action
casper-login              exfat2img               ippeveprinter            mkdosfs                    pptpsetup                  tune2fs                   wpa_cli
casper-new-muid            exfatlabel              ippeveprinter            mkfs                       prime-offload              tune2fs                   wpa_supplicant
casper-snapshot           faillock                 ippeveprinter            mkfs                       prime-switch                ub-device-create          xfs_admin
casper-stop               fattable                 ippeveprinter            mkfs                       pvchange                    u-d-c-print-pci-ids      xfs_bmap
```

This contains all the basic commands used by all users.

/sbin

```
ansh@diazoLaptop:/bin$ cd /sbin
ansh@diazoLaptop:/sbin$ ls
aa-load                  cups-gennpupdate       getty                    jfs_mkfs                 mount.fuse                route                    update-mime
aa-remove-unknown        cupsreject              getweb                   jfs_tune                 mount.fuse3               rsyslogd                update-passwd
aa-status                dcb                     gnome-menus-blacklist  kbdrate                  mount.fountfs-3g          rtacct                  update-pciids
aa-teardown              debugfs                 groupadd                 kerneloops                mount.ntfs                 rtcwake                  update-rc.d
accessdb                 debugfs_reiserfs        groupdel                 key_dns_resolver         mount.ntfs-3g             rtkitctl                update-secureboot-policy
addgnupghome             debugreiserfs           groupmeas                killall5                  mount.smb3                 rtmoo                    update-shells
addgroup                 delgroup                groupmod                 kpartx                    nameif                     runlevel                 update-smart-drivedb
add-shell                 deluser                 grpck                    ldattach                  netplan                    runuser                  upgrade-from-grub-legacy
adduser                  devlink                 grpconv                 ldconfig                  newusers                    samba_kcc                 usb_modeswitch
agetty                   dhclient                grub-bios-setup          lightdm                     nftl_osf                    select-default-ispell    usb_modeswitch_dispatcher
alsa                     dhclient-script         grub-install             lightdm-session            nft                          select-default-wordlist  usbmuxd
alsabat-test             dhcpcd                  grub-macbless            locale-gen                 nologin                     sensors-detect           useradd
alsactl                  dkms                     grub-mkconfig            logrotate                  ntfsclone                   service                  userdel
alsa-info                dmcc                      grub-mkdevicemap         logsave                     ntfscl                       setcap                    usermod
anacron                  dmeventd                grub-probe                lpc                        ntfslabel                   setvesablang             uuid
apparmor_parser          dmidecode               grub-reboot              lpadmin                    ntfsresize                  setvtrgb                  validlocale
apparmor_status          dmraid-activate         grub-set-default          lpmove                     ntfsundelete                sfdisk                   vcsttime
apt                        dmssetup                guest-account             lsdmod                     on_ac_power                 sgdisk                    vdpd
aptk                      dmsstats                halt                       lsmoe                      openvpn                      shadowconfig              veritysetup
arp                       dnsmaq                  hdparm                    lspcacia                   ownership                    shutdown                  vgcfgbackup
arpd                     dosfsck                 iconvconfig               lscnecia                   pam-auth-update             slattach                  vgcfgrestore
arptables                 dosfslabel              ifconfig                  lsdformat                  pam-extrausers_update      slick-greeter             vgcchange
arptables-nft             dpkg-preconfigure       init                       lchance                    pam_extrausers_update      smartctl                  vgcck
arptables-nft-restore     dpkg-reconfigure        insmod                    lcconvert                  pam_getenv                  smartd                     vgcconvert
arptables-nft-save        dumpe2fs                installkernel             lcreate                     pam_namespace_helper        start-stop-daemon         vgcreate
arptables-restore         dump.exfat               install-sgmlcatalog       ldvisplay                  pam_timestamp_check         sudo_logsrvd              vgdisplay
arptables-save            e2freefrag              integritysetup            lexend                       pam_timestam_check          sudo_sendlog              vgexport
aspell-autobuildhash      e2fsck                   invoke-rc.d               lvm                         parted                      sulogin                    vgextend
avahi-autoipd             e2image                 ip                         lvmconfig                  partprobe                   swaplabel                 vgimport
avahi-daemon              e2label                 ip6tables                 lvmformat                  pccardctl                   swapoff                    vgmportclone
badblocks                 e2mmpstatus             ip6tables-apply           lvmmdiskscan               pivot_root                  swapon                     vgmmerge
biosdecode                e2scrub                 ip6tables-legacy          lvmplld                    plipconfig                  switch_root                vgmnodes
blkdeactivate             e2scrub_all              ip6tables-legacy-restore  lvmsadc                     plocate-build               sync-avaiable             vgreduce
blkdiscard                e2undo                  ip6tables-legacy-save     lvmsar                       plymouthd                   sysctl                      vgrename
blkid                     e4crypt                  ip6tables-nft             lyreduce                    pm-hibernate                 system-tools-backends      vgs
blkzone                   e4defrag                ip6tables-nft-restore     lyrename                     pm-powersave                tc                           vgsan
blockdev                  ebtabels                 ip6tables-nft-save        lvresize                     pm-suspend                   tcptraceroute              vgsplit
bluetoothd                ebtabels-nft            ip6tables-restore         lvs                          pm-suspend-hybrid            tcptraceroute.db          vigr
bpftool                   ebtabels-nft-restore    ip6tables-restore-translate  lvscan                       poweroff                      thermald                    vlnv
bridge                     ebtabels-nft-save       ip6tables-restore         mksasl-cert                 pppd                         tipc                        visudo
brltty                    ebtabels-restore         ip6tables-save            mdadm                       pppoe-discovery              tipc                        vpddecode
brltty-setup              ebtabels-save           ip6tables-restore         mdmon                       pppstats                     traceroute                  wipefs
casph                      ebtabels-translate      ippevepcl                 mii-tool                   pptp                          tune2fs                    wpa_action
casper-getty              exfatimg                 ippeveprinter             mkdosfs                     pptpsetip                    tune.exfat                 wpa_cli
casper-login              exfatlabel               ippeveps                  mkfs                       prime-offload                tunefs_reiserfs            wpa_supplicant
casper-new-uid             faillock                  iptables                  mkfs.bfs                    prime-switch                  ub-device-create           xfs_admin
casper-snapshot           fatlabel                 iptables-apply            mksfs                       pvchange                      u-d-c-print-pci-ids        xfs_bmap
casper-stop               fatlabel                 iptables-apply            mksfs                       pvchange                      u-d-c-print-pci-ids        xfs_bmap
```

Contains system admin commands used by root user comparatively lesser than /bin.

/home

```
ansh@diazoLaptop:~$ cd /home
ansh@diazoLaptop:/home$ ls -l
total 8
drwxr-x--- 23 ansh ansh 4096 Jun  8 13:36 ansh
drwxr-x--- 30 diazo diazo 4096 Jun  8 12:31 diazo
```

Learned that it contains users existing in the linux and each directory containing personal files and directory of respective users. Doing cd to the respective folder reaches us to home where our terminal begins. (Note: doing just “cd “ and “cd home/current_user” reaches same destination)

```
ansh@diazoLaptop:/home$ cd diazo
bash: cd: diazo: Permission denied
ansh@diazoLaptop:/home$ su diazo
Password:
diazo@diazoLaptop:/home$ ls
ansh diazo
diazo@diazoLaptop:/home$ cd diazo
diazo@diazoLaptop:~$ ls
Desktop Documents Downloads Learning Music Pictures Public Templates Videos
diazo@diazoLaptop:~$
```

Learned that when trying to access the files of another user permission is denied until we use the switch user command and enter password.

/etc

```

ansh@diazolaptop:~$ cd /etc
ansh@diazolaptop:~$ etcs ls -la
.                  cron.d          gai.conf          inxi.conf         machine-id        pan.conf          rmt              systemd
..                 cron.daily       gdm              lpp-usb          magic             pan.d             rpc              terminfo
adduser.conf       cron.hourly     gdm.conf         iproute2         magic.mime        papersize         rsyslog.conf     thermal
adjtime           cron.monthly    ghostscript      issue            mailcap           passwd           rsyslog.d        timeshift
alsa              cron.ontab      glvnd            issue.net        mailcap.order     passwd           samba            timezone
alternatives       cron.weekly     gnome            java              manpath.conf     pci             sane.d           timidity
anacron            cron.yearly     gnome-app-install kernel            mddm             perl             security         ttfconfig.d
app-conf           cryptsetup-initramfs gnutils          kernel           mtime.types      pk1              selinux          ucr.conf
apm               cups            groffng.rc       kerneloops.conf mke2fs.conf      plymouth         sensors3.conf    udev
apparmor           cupselppers     groff            keyutils         ModemManager     pm              sensors.d        udisks2
apparmor.d         dbus-1          group           ld.so.cache      modprobe.d        pnp2ppa.conf     services        urfw
apt               debconf         grub.d           ld.so.conf        modules           polkit-1         sgml             updatedb.conf
avahi              debconf.conf    grub.d           ld.so.conf.d      modules-load.d    ppm             shadow           update-motd.d
bash.bashrc        debian.version  gshadow          libao.conf       mdt              prime-discrete   shells           UPower
bash_completion    debuginfo      gshadow          libaudit.conf    mtab              profile.d         skel             usb_modeswitch
bash_completion.d  default        gss              libblockdev       mttools.conf      protocols        smartd.conf      usb_modeswitch.d
bindresolv.conf    deluser.conf   gtk-2.0          libl3             nanorc            python3           smartmontools    vconsole.conf
bluetooth          dhcp            guest-session    libpaper.d        netconfig         rc0.d             speech-dispatcher vdpau_wrapper.cfg
brlapi.key         dhcpcd.conf     gufw             libreoffice       network           rc1.d             ssh              vtrgb
brlty.conf         dictionaries-common hdpam.conf       lightdm           networkd-dispatcher rc2.d             ssl              vulkan
ca-certificates    dirmngr        host.conf        linuxmint         NetworkManager   rc3.d             sudo             wgetrc
ca-certificates.conf dpkg            hostname         locale.alias      networks          rc4.d             sudo.d           xdg
casper.conf        emacscap       hosts            locale.gen        newt              rc5.d             suders           xml
catdoc             environment     hosts.allow      localtime         nftables.conf    rc6.d             sudoers          zsh_command_not_found
chatscripts        ethtypes       hosts.deny       logcheck          nfswitch.conf    rc7.d             sudoers.d        zsh
cifs-utils         fonts          ifplugd          login.defs        openal            rc8.d             sudoers.vrf     zsh
color              fonts          ImageMagick-6    logrotate.d       OpenCL            rc9.d             sudoers.vrf     zsh
console-setup      freintd.conf   init             logrotate.d       openvpn           request-key.conf  sudo.logsvd.conf zsh
cramlib            frib           inittab          lsb_release       openssl           request-key.d     sudoers.vrf     zsh
credstore          fuse.conf      initscripts-tools lrtrace.conf      resolv.conf       resolv.conf       sysctl.conf      zsh
credstore.encrypted fwupd          inputrc          lvm               os-release        resolv.conf       sysctl.d         zsh
ansh@diazolaptop:~$ etcs cat hosts
127.0.0.1          localhost
127.0.1.1          diazolaptop

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe80:: ip6-localhost
fe00:: ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

```

Exploring the /etc folder, it contains system-wide configuration files. Accessed hosts to learn what it contains.

```
/var
```

```

ansh@diazoLaptop:~$ cd /var
ansh@diazoLaptop:/var$ ls -la
. . . backups cache lib local lock log mail opt run spool [redacted] .updated
ansh@diazoLaptop:/var$ cd
ansh@diazoLaptop:~$ cd /var
ansh@diazoLaptop:/var$ ls
backups cache lib local lock log mail opt run spool [redacted]
ansh@diazoLaptop:/var$ cd log
ansh@diazoLaptop:/var/log$ ls
alternatives.log  btmo           installer      openvpn        timeshift      Xorg.0.log.old
apt               cups           dpkg.log       journal         ubuntu-system-adjustments-adjust-grub-title.log   Xorg.1.log
auth.log          cups-browsed   faillog        kern.log        private        ubuntu-system-adjustments-start.log              Xorg.1.log.old
auth.log.1        dmesg          fontconfig.log kern.log.1      README         ubuntu-system-adjustments-stop.log               Xorg.2.log
boot.log          dmesg.0        gpu-manager.log lastlog         samba           ufw.log
boot.log.1        dmesg.1.gz     gpu-manager-switch.log lightdm         speech-dispatcher  ufw.log.1
boot.log.2        dmesg.2.gz     gufw.log       mntsystem.log  syslog         wtmp
bootstrap.log     dmesg.3.gz     hp             mintsytem.timestamps syslog.1        Xorg.0.log

ansh@diazoLaptop:/var/log$ cat README
You are looking for the traditional text log files in /var/log, and they are gone?

Here's an explanation on what's going on:

You are running a systemd-based OS where traditional syslog has been replaced with the Journal. The journal stores the same (and more) information as classic syslog. To make use of the journal and access the collected log data simply invoke "journalctl", which will output the logs in the identical text-based format the syslog files in /var/log used to be. For further details, please refer to journalctl(1).

Alternatively, consider installing one of the traditional syslog implementations available for your distribution, which will generate the classic log files for you. Syslog implementations such as syslog-ng or rsyslog may be installed side-by-side with the journal and will continue to function the way they always did.

Thank you!

Further reading:
man:journalctl(1)
man:systemd-journald.service(8)
man:journald.conf(5)
https://0pointer.de/blog/projects/the-journal.html

```

While accessing var learnt that it stores files that change frequently such as logs and cache.

Out of curiosity I explored what a log file looks like and found out some files couldn't be accessed while some could be accessed.

/usr

```
ansh@diazoLaptop:~$ cd /usr
ansh@diazoLaptop:/usr$ ls -la
. . bin games include lib lib64 libexec local sbin share src
ansh@diazoLaptop:/usr$ cd bin
ansh@diazoLaptop:/usr/bin$ ls -la
. .
. .
. . foo2lava-wrapper mintBackup sha384sum
. . foo2oak mintchat sha512sum
. . foo2oak-wrapper mintchat-launch shasum
7z foo2qpdfL mint-drivers showconsolefont
7za foo2qpdfL-wrapper mintdrivers showkey
7zr foo2sfx mintdrivers-add-live-media showrgb
aa-enabled foo2sfx-wrapper mintdrivers-load-broadcom-modules shred
aa-exec foo2sfxq mintdrivers-remove-live-media shuf
aa-features-abi foo2sfxq-wrapper mintinstall simple-scan
aconnect foo2zjs mintinstall-fp-handler size
acpidbg foo2zjs-icc2ps mintinstall-update-pkgcache skill
add-apt-repository foo2zjs-pstps mint-iso-verify slabtop
addpart foo2zjs-wrapper mintlocale slack
addr2line foomatic-rip mintlocale-im sleep
add-remove-locales foointd-delete mintmenu click-propter-check-hidpi
```

Contains user programs and libraries. Learned that /usr/bin is accessed from /bin before. That is /bin is a symbolic link to /usr/bin.

/boot

```
ansh@diazoLaptop:~$ cd /boot
ansh@diazoLaptop:/boot$ ls -la
. . config-6.14.0-37-generic efi initrd.img initrd.img-6.17.0-35-generic System.map-6.14.0-37-generic vmlinuz vmlinuz-6.17.0-35-generic
. . config-6.17.0-35-generic grub initrd.img-6.14.0-37-generic initrd.img.old System.map-6.17.0-35-generic vmlinuz-6.14.0-37-generic vmlinuz.old
```

Contains everything required for linux to boot.

Linux commands:

Print working directory - pwd

```
ansh@diazoLaptop:~$ pwd
/home/ansh
```

Initially testing ls and its modifiers -l and -a:

```
ansh@diazoLaptop:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
ansh@diazoLaptop:~$ ls -a
.          .bash_logout  .config  Desktop  Documents  .gtkrc-2.0  .linuxmint  Music  .nvidia-settings-rc  .profile  .sudo_as_admin_successful  .thunderbird  .vscode-shared  .xsession-errors.d
..         .bash_history .cache    .dmrc    Downloads  .gtkrc-xfce .local      .mozilla  Pictures  .ssh      .templates  .vscode      .xsession-errors
ansh@diazoLaptop:~$ ls -l
total 32
drwxr-xr-x 2 ansh ansh 4096 Jun  7 13:22 Desktop
drwxr-xr-x 2 ansh ansh 4096 Jun  7 13:22 Documents
drwxr-xr-x 2 ansh ansh 4096 Jun  7 13:22 Downloads
drwxr-xr-x 2 ansh ansh 4096 Jun  7 13:22 Music
drwxr-xr-x 2 ansh ansh 4096 Jun  7 13:22 Pictures
drwxr-xr-x 2 ansh ansh 4096 Jun  7 13:22 Public
drwxr-xr-x 2 ansh ansh 4096 Jun  7 13:22 Templates
drwxr-xr-x 2 ansh ansh 4096 Jun  7 13:22 Videos
```

Making directory and file for testing other commands:

```
ansh@diazoLaptop:~$ mkdir Task1
ansh@diazoLaptop:~$ cd Task1
ansh@diazoLaptop:~/Task1$ nano file1.task
ansh@diazoLaptop:~/Task1$ cd ..
ansh@diazoLaptop:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Task1  Templates  Videos
```

tree

Sudo apt install tree for accessing tree and the changing to Task1 Folder

```
ansh@diazoLaptop:~$ tree
.
├── Desktop
├── Documents
│   └── Learning
├── Downloads
│   ├── Linux Filesystem Hierarchy and Essential Commands.pdf
│   └── prfile.jpeg
├── Music
├── Pictures
├── Public
├── Task1
│   └── file1.task
├── Templates
└── Videos

10 directories, 4 files
```

cp

```
ansh@diazoLaptop:~/Task1$ ls
file1.task
ansh@diazoLaptop:~/Task1$ cd
ansh@diazoLaptop:~$ touch file2.hello
ansh@diazoLaptop:~$ ls
Desktop Documents Downloads file2.hello Music Pictures Public Task1 Templates Videos
ansh@diazoLaptop:~$ cp file2.hello Task1
ansh@diazoLaptop:~$ ls /Task1
ls: cannot access '/Task1': No such file or directory
ansh@diazoLaptop:~$ ls Task1
file1.task file2.hello

ansh@diazoLaptop:~$ ls
Desktop Documents Downloads file2.hello Music Pictures Public Task1 Templates Videos
ansh@diazoLaptop:~$ cp -r Task1/ TaskCopy
ansh@diazoLaptop:~$ cd TaskCopy
ansh@diazoLaptop:~/TaskCopy$ ls
file1.task file2.hello
```

Recursively copying all files from Task1 to TaskCopy.

mv

```
ansh@diazoLaptop:~$ ls
Desktop Documents Downloads file2.hello Music Pictures Public Task1 TaskCopy Templates Videos
ansh@diazoLaptop:~$ cd Task1
ansh@diazoLaptop:~/Task1$ ls
file1.task file2.hello
ansh@diazoLaptop:~/Task1$ mv file1.task ../
ansh@diazoLaptop:~/Task1$ cd
ansh@diazoLaptop:~$ ls
Desktop Documents Downloads file1.task file2.hello Music Pictures Public Task1 TaskCopy Templates Videos
ansh@diazoLaptop:~$ cd Task1
ansh@diazoLaptop:~/Task1$ ls
file2.hello
ansh@diazoLaptop:~/Task1$ cd ..
ansh@diazoLaptop:~$ mv file1.task Task1
ansh@diazoLaptop:~$ cd Task1
ansh@diazoLaptop:~/Task1$ ls
file1.task file2.hello
ansh@diazoLaptop:~/Task1$ mv file2.hello file2.task
ansh@diazoLaptop:~/Task1$ ls
file1.task file2.task
```

Used mv to move files as well as rename files.

rm

```
ansh@diazoLaptop:~$ ls
Desktop Documents Downloads file2.hello Music Pictures Public Task1 Templates Videos
ansh@diazoLaptop:~$ rm file2.hello
ansh@diazoLaptop:~$ ls
Desktop Documents Downloads Music Pictures Public Task1 Templates Videos
```

rm -r

```
ansh@diazoLaptop:~$ rm -r Task1
ansh@diazoLaptop:~$ ls Task1
ls: cannot access 'Task1': No such file or directory
ansh@diazoLaptop:~$ ls
Desktop Documents Downloads Music Pictures Public Task2 TaskCopy Templates Videos
ansh@diazoLaptop:~$
```

When directory has files we can remove it as well as folder with rm -r.

find

```
ansh@diazoLaptop:~/Task1$ ls
file1.task file2.hello file2.task
ansh@diazoLaptop:~/Task1$ cd
ansh@diazoLaptop:~$ find . -name "file2.hello"
./Task1/file2.hello
./TaskCopy/file2.hello
ansh@diazoLaptop:~$ find . -name "*.log"
../local/share/gvfs-metadata/home-e448273b.log
../config/Slack/IndexedDB/https_app.slack.com_0.indexeddb.leveldb/000003.log
../config/Slack/Local Storage/leveldb/000044.log
../config/Slack/Session Storage/000003.log
../config/Slack/logs/default/webapp-console.log
../config/Slack/logs/default/webapp-service-worker-console.log
../config/Slack/logs/default/browser.log
../config/Slack/Service Worker/Database/000003.log
../config/mozilla/firefox/Crash Reports/crash_helper_server.log
../config/Code/Local Storage/leveldb/000003.log
../config/Code/Session Storage/000003.log
../config/Code/logs/20260608T094553/main.log
../config/Code/logs/20260608T094553/sharedprocess.log
../config/Code/logs/20260608T094553/tunnelHostService.log
../config/Code/logs/20260608T094553/window1/views.log
../config/Code/logs/20260608T094553/window1/output_20260608T094607/agentSessionsOutput.log
../config/Code/logs/20260608T094553/window1/output_20260608T094607/tasks.log
../config/Code/logs/20260608T094553/window1/notebook.rendering.log
../config/Code/logs/20260608T094553/window1/network.log
../config/Code/logs/20260608T094553/window1/exthost/vscode.github-authentication/GitHub Authentication.log
../config/Code/logs/20260608T094553/window1/exthost/extHostTelemetry.log
../config/Code/logs/20260608T094553/window1/exthost/vscode.git/Git.log
../config/Code/logs/20260608T094553/window1/exthost/vscode.json-language-features/JSON Language Server.log
../config/Code/logs/20260608T094553/window1/exthost/exthost.log
../config/Code/logs/20260608T094553/window1/exthost/vscode.github/GitHub.log
../config/Code/logs/20260608T094553/window1/renderer.log
```

Linux User, Group and Permission

useradd and passwd

```
ansh@diazoLaptop:~$ sudo useradd Guest
ansh@diazoLaptop:~$ sudo passwd Guest
New password:
Retype new password:
passwd: password updated successfully
```

groupadd

```
ansh@diazoLaptop:~$ sudo groupadd fellowship
```

Added a group named fellowship.

```
ansh@diazoLaptop:~$ sudo usermod -g fellowship Guest
ansh@diazoLaptop:~$ id Guest
uid=1002(Guest) gid=1003(fellowship) groups=1003(fellowship)
ansh@diazoLaptop:~$ sudo usermod -aG fellowship ansh
ansh@diazoLaptop:~$ id ansh
uid=1001(ansh) gid=1001(ansh) groups=1001(ansh),27(sudo),1003(fellowship)
```

Adding a group then adding users to the group. Learned that -g means setting as primary group and also learnt about supplementary group in linux created with -aG where -a means append and -G means supplementary group. (supplementary group doesn't default set file created by user into group unlike primary)

chown

Changing ownership of files that were created earlier and learning about commands like chown.

```
ansh@diazoLaptop:~/Task1$ ls
file1.task file2.hello file2.task
ansh@diazoLaptop:~/Task1$ ls -l
total 4
-rw-rw-r-- 1 ansh ansh 44 Jun  8 19:08 file1.task
-rw-rw-r-- 1 ansh ansh  0 Jun  8 19:11 file2.hello
-rw-rw-r-- 1 ansh ansh  0 Jun  8 19:11 file2.task
ansh@diazoLaptop:~/Task1$ sudo chown ansh file1.task
[sudo] password for ansh:
ansh@diazoLaptop:~/Task1$ sudo chown Guest file2.hello
ansh@diazoLaptop:~/Task1$ ls -l
total 4
-rw-rw-r-- 1 ansh ansh 44 Jun  8 19:08 file1.task
-rw-rw-r-- 1 Guest ansh  0 Jun  8 19:11 file2.hello
-rw-rw-r-- 1 ansh ansh  0 Jun  8 19:11 file2.task
ansh@diazoLaptop:~/Task1$ sudo chown Guest:fellowship file2.hello
ansh@diazoLaptop:~/Task1$ ls -l
total 4
-rw-rw-r-- 1 ansh ansh 44 Jun  8 19:08 file1.task
-rw-rw-r-- 1 Guest fellowship 0 Jun  8 19:11 file2.hello
-rw-rw-r-- 1 ansh ansh  0 Jun  8 19:11 file2.task
ansh@diazoLaptop:~/Task1$ sudo chown ansh:fellowship file1.task
ansh@diazoLaptop:~/Task1$ ls -l
total 4
-rw-rw-r-- 1 ansh fellowship 44 Jun  8 19:08 file1.task
-rw-rw-r-- 1 Guest fellowship 0 Jun  8 19:11 file2.hello
-rw-rw-r-- 1 ansh ansh  0 Jun  8 19:11 file2.task
```


chmod

In chmod the permission can be changed with the help of values.

Read (r) -> 4

Write (w) -> 2

Execute (x) -> 1

So if we need rw we write 6 (4+2)

Example: -rw-r--r- : 644

If we have to make readonly by the owner : 400

```
ansh@diazoLaptop:~/Task1$ ls -l
total 4
-rw-rw-r-- 1 ansh  fellowship 44 Jun  8 19:08 file1.task
-rw-rw-r-- 1 Guest fellowship  0 Jun  8 19:11 file2.hello
-rw-rw-r-- 1 ansh  ansh        0 Jun  8 19:11 file2.task
ansh@diazoLaptop:~/Task1$ chmod 644 file1.task
ansh@diazoLaptop:~/Task1$ ls -l
total 4
-rw-r--r-- 1 ansh  fellowship 44 Jun  8 19:08 file1.task
-rw-rw-r-- 1 Guest fellowship  0 Jun  8 19:11 file2.hello
-rw-rw-r-- 1 ansh  ansh        0 Jun  8 19:11 file2.task
```

Git & Version Control

Creating folder initializing git and adding it using git add

```
ansh@diazoLaptop:~/Task1$ mkdir git-practice
ansh@diazoLaptop:~/Task1$ cd git-practice
ansh@diazoLaptop:~/Task1/git-practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/ansh/Task1/git-practice/.git/
ansh@diazoLaptop:~/Task1/git-practice$ git branch -M main
ansh@diazoLaptop:~/Task1/git-practice$ touch textfile.txt
ansh@diazoLaptop:~/Task1/git-practice$ git add textfile.txt
ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   textfile.txt
```

git commit

```
ansh@diazoLaptop:~/Task1/git-practice$ git commit -m "docs:add textfile.txt"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'ansh@diazoLaptop.(none)')
ansh@diazoLaptop:~/Task1/git-practice$ git config --global user.email "anshshrestha15@gmail.com"
ansh@diazoLaptop:~/Task1/git-practice$ git config --global user.name "D1AZ0W"
ansh@diazoLaptop:~/Task1/git-practice$ git commit -m "docs:add textfile.txt"
[main (root-commit) 5d27b9a] docs:add textfile.txt
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 textfile.txt
ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch main
nothing to commit, working tree clean
ansh@diazoLaptop:~/Task1/git-practice$ git log
commit 5d27b9a60ccc49866d37e1754be480eb7c7eb3be0 (HEAD -> main)
Author: D1AZ0W <anshshrestha15@gmail.com>
Date:   Mon Jun 8 20:57:05 2026 +0545

    docs:add textfile.txt
```

Creating branches

```
ansh@diazoLaptop:~/Task1/git-practice$ git checkout -b development
Switched to a new branch 'development'
ansh@diazoLaptop:~/Task1/git-practice$ git branch
* development
  main

ansh@diazoLaptop:~/Task1/git-practice$ git checkout -b docs/more-information
Switched to a new branch 'docs/more-information'
ansh@diazoLaptop:~/Task1/git-practice$ git branch
  development
* docs/more-information
  main
ansh@diazoLaptop:~/Task1/git-practice$ nano moreInformation.txt
ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch docs/more-information
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    moreInformation.txt

nothing added to commit but untracked files present (use "git add" to track)
ansh@diazoLaptop:~/Task1/git-practice$ git add moreInformation.txt
ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch docs/more-information
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   moreInformation.txt

ansh@diazoLaptop:~/Task1/git-practice$ git commit "docs: moreInformation is added"
error: pathspec 'docs: moreInformation is added' did not match any file(s) known to git
ansh@diazoLaptop:~/Task1/git-practice$ git commit -m "docs: moreInformation is added"
[docs/more-information b22f6d4] docs: moreInformation is added
1 file changed, 1 insertion(+)
create mode 100644 moreInformation.txt
ansh@diazoLaptop:~/Task1/git-practice$ git log
commit b22f6d4c86110c087d31cb4d1caf97df59e5319a (HEAD -> docs/more-information)
Author: D1AZ0W <anshshrestha15@gmail.com>
Date:   Mon Jun 8 21:06:43 2026 +0545

    docs: moreInformation is added

commit 5d27b9a60cc49866d37e1754be480eb7c7eb3be0 (main, development)
Author: D1AZ0W <anshshrestha15@gmail.com>
Date:   Mon Jun 8 20:57:05 2026 +0545

    docs:add textfile.txt
```

Added another branch named docs/more-information after learning (<https://pullpanda.io/blog/git-branch-naming-conventions-best-practices>) and added moreInformation.txt and committed to that branch.

git merge

checkout to development and merge development and docs/more-information

```
ansh@diazoLaptop:~/Task1/git-practices$ git branch
  development
* docs/more-information
  main
ansh@diazoLaptop:~/Task1/git-practices$ git checkout development
Switched to branch 'development'
ansh@diazoLaptop:~/Task1/git-practices$ git branch
* development
  docs/more-information
  main
ansh@diazoLaptop:~/Task1/git-practices$ git merge docs/more-information
Updating 5d27b9a..b22f6d4
Fast-forward
 moreInformation.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 moreInformation.txt
```

git merge conflict

```
• ansh@diazoLaptop:~/Task1/git-practice$ git branch
  development
  docs/more-information
* main
• ansh@diazoLaptop:~/Task1/git-practice$ git checkout development
Switched to branch 'development'
• ansh@diazoLaptop:~/Task1/git-practice$ git checkout -b feat/userInfo
Switched to a new branch 'feat/userInfo'
• ansh@diazoLaptop:~/Task1/git-practice$ git branch
  development
  docs/more-information
* feat/userInfo
  main
• ansh@diazoLaptop:~/Task1/git-practice$ nano userInfo.txt
• ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch feat/userInfo
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   moreInformation.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        userInfo.txt

no changes added to commit (use "git add" and/or "git commit -a")
• ansh@diazoLaptop:~/Task1/git-practice$ git add moreInformation.txt userInfo.txt
```

adding a new branch where we add new userInfo.txt and also made changes to moreInformation.txt.

```
• ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch feat/userInfo
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   moreInformation.txt
        new file:   userInfo.txt

• ansh@diazoLaptop:~/Task1/git-practice$ git commit -m "feat: added userInfo"
[feat/userInfo ecaf4c7] feat: added userInfo
2 files changed, 3 insertions(+)
create mode 100644 userInfo.txt
```

Then we commit the change in branch feat/userInfo.

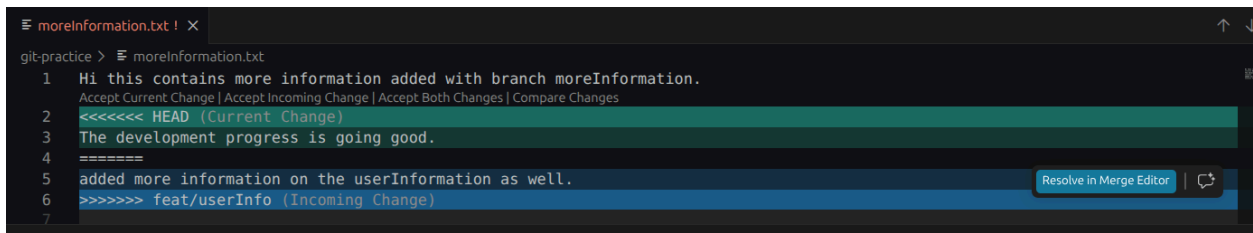
```

• ansh@diazoLaptop:~/Task1/git-practice$ git branch
  development
  docs/more-information
  * feat/userInfo
  main
• ansh@diazoLaptop:~/Task1/git-practice$ git checkout development
Switched to branch 'development'
• ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch development
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   moreInformation.txt

no changes added to commit (use "git add" and/or "git commit -a")
• ansh@diazoLaptop:~/Task1/git-practice$ git add moreInformation.txt
• ansh@diazoLaptop:~/Task1/git-practice$ git branch
  * development
  docs/more-information
  feat/userInfo
  main

```

Then we checkout to development and again bring changes to moreInformation.txt.



```

moreInformation.txt ! X
git-practice > moreInformation.txt
1 Hi this contains more information added with branch moreInformation.
  Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
2 <<<<<< HEAD (Current Change)
3 The development progress is going good.
4 =====
5 added more information on the userInfo as well.
6 >>>>>> feat/userInfo (Incoming Change)
7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - git-practice
• ansh@diazoLaptop:~/Task1/git-practice$ git branch
  * development
  docs/more-information
  feat/userInfo
  main
• ansh@diazoLaptop:~/Task1/git-practice$ git merge feat/userInfo
Updating b22f6d4..ecaf4c7
error: Your local changes to the following files would be overwritten by merge:
  moreInformation.txt
Please commit your changes or stash them before you merge.
Aborting
• ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch development
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   moreInformation.txt
• ansh@diazoLaptop:~/Task1/git-practice$ git commit "docs: moreInformation updated"
error: pathspec 'docs: moreInformation updated' did not match any file(s) known to git
• ansh@diazoLaptop:~/Task1/git-practice$ git commit -m "docs: moreInformation updated"
[development 9815634] docs: moreInformation updated
 1 file changed, 1 insertion(+)
• ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch development
nothing to commit, working tree clean
• ansh@diazoLaptop:~/Task1/git-practice$ git merge feat/userInfo
Auto-merging moreInformation.txt
CONFLICT (content): Merge conflict in moreInformation.txt
Automatic merge failed; fix conflicts and then commit the result.

```

When committing the change in development branch after changing moreInformation.txt and merging with feat/userInfo there arises a git merge conflict. To solve this issue we need to bring changes to development branch only with the help of branches and not directly in this case moreInformation can be added from another branch then can be reviewed and merged again.

git rebase

Trying git rebase and exploring in same files and situation.

```
• ansh@diazoLaptop:~/Task1/git-practice$ git checkout feat/userInfo
Switched to branch 'feat/userInfo'
• ansh@diazoLaptop:~/Task1/git-practice$ git rebase development
Auto-merging moreInformation.txt
CONFLICT (content): Merge conflict in moreInformation.txt
error: could not apply ecaf4c7... feat: added userInfo
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
Could not apply ecaf4c7... feat: added userInfo
```

This still causes conflict because both commits modified moreInformation.txt.

```
GNU nano 7.2 moreInformation.txt
Hi this contains more information added with branch moreInformation.
<<<<<< HEAD
The development progress is going good.
=====
added more information on the userInfo as well.
>>>>>> ecaf4c7 (feat: added userInfo)
```

Manually removing conflict only keeping from feat:added userInfo

```
• ansh@diazoLaptop:~/Task1/git-practice$ nano moreInformation.txt
• ansh@diazoLaptop:~/Task1/git-practice$ git add moreInformation.txt
• ansh@diazoLaptop:~/Task1/git-practice$ git rebase --continue
[detached HEAD 3e671a1] feat: added userInfo
2 files changed, 3 insertions(+), 1 deletion(-)
create mode 100644 userInfo.txt
Successfully rebased and updated refs/heads/feat/userInfo.
```

```
• ansh@diazoLaptop:~/Task1/git-practice$ git checkout development
Switched to branch 'development'
• ansh@diazoLaptop:~/Task1/git-practice$ git status
On branch development
nothing to commit, working tree clean
• ansh@diazoLaptop:~/Task1/git-practice$ git merge feat/userInfo
Updating 9815634..3e671a1
Fast-forward
 moreInformation.txt | 2 +-
 userInfo.txt        | 2 ++
2 files changed, 3 insertions(+), 1 deletion(-)
create mode 100644 userInfo.txt
```

Then we can merge in development too.


```

• ansh@diazoLaptop:~/Task1/git-practice$ git checkout -b feat/login
Switched to a new branch 'feat/login'
• ansh@diazoLaptop:~/Task1/git-practice$ git branch
  development
  docs/more-information
* feat/login
  feat/userInfo
  main
• ansh@diazoLaptop:~/Task1/git-practice$ nano login.info
Ⓢ ansh@diazoLaptop:~/Task1/git-practice$ git commit -m "feat: added login"
On branch feat/login
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    login.info

nothing added to commit but untracked files present (use "git add" to track)
• ansh@diazoLaptop:~/Task1/git-practice$ git add login.info
• ansh@diazoLaptop:~/Task1/git-practice$ git commit -m "feat: added login"
[feat/login d3f29b0] feat: added login
 1 file changed, 1 insertion(+)
 create mode 100644 login.info
• ansh@diazoLaptop:~/Task1/git-practice$ git checkout main
Switched to branch 'main'
• ansh@diazoLaptop:~/Task1/git-practice$ git checkout development
Switched to branch 'development'
• ansh@diazoLaptop:~/Task1/git-practice$ git rebase feat/login

```

Practiced rebase again by rebasing development with feat/login

```

• ansh@diazoLaptop:~/Task1/git-practice$ git merge development
Updating b22f6d4..ecdb87a
Fast-forward
 login.info          | 1 +
 moreInformation.txt | 1 +
 userInfo.txt        | 2 ++
 3 files changed, 4 insertions(+)
 create mode 100644 login.info
 create mode 100644 userInfo.txt

```

Final merge with main and development.